

MINI PROJECT – DAY 9 OF 3

Hospital Data Pipeline

Phase 2: Cleaning, Transformation, Enrichment

STEP 1 - Linux pre-check

From hospital_pipeline directory:

1. Confirm all three raw CSV files exist.

```
ls -lh raw/
```

2. Check for any blank lines in each file.

```
grep -c "^$" raw/patients.csv
```

3. Confirm row counts match Day 8.

```
wc -l raw/*.csv
```

```
prashant@DESKTOP-B9KE2N7:/mnt/d/kode-x notes/python/python_exercise's/day6_project/hospital_pipeline$ ls -lh raw/
total 12K
-rwxrwxrwx 1 prashant prashant 465 Mar 23 22:23 appointments.csv
-rwxrwxrwx 1 prashant prashant 206 Mar 23 22:22 doctors.csv
-rwxrwxrwx 1 prashant prashant 256 Mar 23 22:20 patients.csv
-rwxrwxrwx 1 prashant prashant 8.8K Mar 24 11:11 profiler.ipynb
prashant@DESKTOP-B9KE2N7:/mnt/d/kode-x notes/python/python_exercise's/day6_project/hospital_pipeline$ grep -c "^$" raw/patients.csv
0
prashant@DESKTOP-B9KE2N7:/mnt/d/kode-x notes/python/python_exercise's/day6_project/hospital_pipeline$ grep -c "^$" raw/appointments.c
sv
0
prashant@DESKTOP-B9KE2N7:/mnt/d/kode-x notes/python/python_exercise's/day6_project/hospital_pipeline$ grep -c "^$" raw/patients.csv
0
prashant@DESKTOP-B9KE2N7:/mnt/d/kode-x notes/python/python_exercise's/day6_project/hospital_pipeline$ wc -l raw/*.csv
11 raw/appointments.csv
6 raw/doctors.csv
9 raw/patients.csv
26 total
prashant@DESKTOP-B9KE2N7:/mnt/d/kode-x notes/python/python_exercise's/day6_project/hospital_pipeline$
```

STEP 2 - Python: Clean each file

Create: processed/cleaner.py

Task A - Clean patients.csv:

1. Load raw/patients.csv.

```
# Task A - Clean patients.csv:
# 1. Load raw/patients.csv.

import pandas as pd
df = pd.read_csv("patients.csv")
print(df)
```

✓ 0.0s

	patient_id	name	age	blood_type	city
0	1	Raj Kumar	34	O+	Delhi
1	2	Sneha Patel	39	A+	Mumbai
2	3	Tom Williams	46	B-	Chicago
3	4	Fatima Al-Sayed	29	AB+	Dubai
4	5	Liu Yang	24	O-	Beijing
5	6	Ana Gonzalez	56	A-	Madrid
6	7	James Okafor	32	B+	Lagos
7	8	Emily Carter	19	O+	Toronto

2. Strip whitespace from all string columns.

```
# 2. Strip whitespace from all string columns.

df = df.apply(lambda column: column.str.strip() if column.dtype == object else column)
print(df)
```

✓ 0.0s

	patient_id	name	age	blood_type	city
0	1	Raj Kumar	34	O+	Delhi
1	2	Sneha Patel	39	A+	Mumbai
2	3	Tom Williams	46	B-	Chicago
3	4	Fatima Al-Sayed	29	AB+	Dubai
4	5	Liu Yang	24	O-	Beijing
5	6	Ana Gonzalez	56	A-	Madrid
6	7	James Okafor	32	B+	Lagos
7	8	Emily Carter	19	O+	Toronto

3. Convert name to Title Case.

```
# 3. Convert name to Title Case.
if "name" in df.columns:
    df["name"] = df["name"].str.title()
print(df)
```

✓ 0.0s

	patient_id	name	age	blood_type	city
0	1	Raj Kumar	34	O+	Delhi
1	2	Sneha Patel	39	A+	Mumbai
2	3	Tom Williams	46	B-	Chicago
3	4	Fatima Al-Sayed	29	AB+	Dubai
4	5	Liu Yang	24	O-	Beijing
5	6	Ana Gonzalez	56	A-	Madrid
6	7	James Okafor	32	B+	Lagos
7	8	Emily Carter	19	O+	Toronto

4. Convert city to Title Case.

```
# 4. Convert city to Title Case.
if "city" in df.columns:
    df["city"] = df["city"].str.title()
    print(df)
```

✓ 0.0s

	patient_id	name	age	blood_type	city
0	1	Raj Kumar	34	O+	Delhi
1	2	Sneha Patel	39	A+	Mumbai
2	3	Tom Williams	46	B-	Chicago
3	4	Fatima Al-Sayed	29	AB+	Dubai
4	5	Liu Yang	24	O-	Beijing
5	6	Ana Gonzalez	56	A-	Madrid
6	7	James Okafor	32	B+	Lagos
7	8	Emily Carter	19	O+	Toronto

5. Check for duplicate patient_ids.

Print count of duplicates.

```
# 5. Check for duplicate patient_ids.
# Print count of duplicates.
|
if "patient_id" in df.columns:
    duplicate_count = df["patient_id"].duplicated().sum()
    print(f"Count of duplicates: {duplicate_count}")
```

✓ 0.0s

Count of duplicates: 0

6. Check for NULL values in any column.

Print a summary.

```
# 6. Check for NULL values in any column.
# Print a summary.
print(df.isnull().sum())
```

✓ 0.0s

```
patient_id    0
name          0
age           0
blood_type    0
city          0
age_group     0
dtype: int64
```

7. Add a column age_group:

age < 30 → "Young"

age < 50 → "Adult"

age >= 50 → "Senior"

```
# 7. Add a column age_group:
#   age < 30 → "Young"
#   age < 50 → "Adult"
#   age >= 50 → "Senior"
'''
def age_band(x):
    if x < 30:
        return "Young"
    elif x < 50:
        return "Adult"
    else:
        return "Senior"

df["age_group"] = df["age"].apply(age_band)
print(df)
'''
import numpy as np
if "age" in df.columns:
    conditions = [
        df["age"] < 30,
        df["age"] < 50,
        df["age"] >= 50
    ]
    choices = ["Young", "Adult", "Senior"]

    df["age_group"] = np.select(conditions, choices, default="Unknown")
    print(df)
```

✓ 0.0s

Python

	patient_id	name	age	blood_type	city	age_group
0	1	Raj Kumar	34	O+	Delhi	Adult
1	2	Sneha Patel	39	A+	Mumbai	Adult
2	3	Tom Williams	46	B-	Chicago	Adult
3	4	Fatima Al-Sayed	29	AB+	Dubai	Young
4	5	Liu Yang	24	O-	Beijing	Young
5	6	Ana Gonzalez	56	A-	Madrid	Senior
6	7	James Okafor	32	B+	Lagos	Adult
7	8	Emily Carter	19	O+	Toronto	Young

8. Save to processed/patients_clean.csv

without index.

```
# 8. Save to processed/patients_clean.csv
# without index
df.to_csv("processed/patients_clean.csv", index=False)
df.head(8)
```

✓ 0.0s

	patient_id	name	age	blood_type	city	age_group
0	1	Raj Kumar	34	O+	Delhi	Adult
1	2	Sneha Patel	39	A+	Mumbai	Adult
2	3	Tom Williams	46	B-	Chicago	Adult
3	4	Fatima Al-Sayed	29	AB+	Dubai	Young
4	5	Liu Yang	24	O-	Beijing	Young
5	6	Ana Gonzalez	56	A-	Madrid	Senior
6	7	James Okafor	32	B+	Lagos	Adult
7	8	Emily Carter	19	O+	Toronto	Young

Task B - Clean doctors.csv:

1. Load raw/doctors.csv.

```
# Task B - Clean doctors.csv:
# 1. Load raw/doctors.csv.
import pandas as pd

df = pd.read_csv("doctors.csv")
df.head()
```

✓ 0.0s

	doctor_id	name	specialty	experience_years
0	1	Dr. Anil Mehta	Cardiology	15
1	2	Dr. Sarah Johnson	Neurology	12
2	3	Dr. Raj Singh	Orthopedics	8
3	4	Dr. Lisa Park	Dermatology	6
4	5	Dr. Carlos Torres	General Medicine	20

2. Strip whitespace from all string columns.

```
# 2. Strip whitespace from all string columns.
df = df.apply(lambda col: col.str.strip() if col.dtype == object else col)
print(df)
```

✓ 0.0s

	doctor_id	name	specialty	experience_years
0	1	Dr. Anil Mehta	Cardiology	15
1	2	Dr. Sarah Johnson	Neurology	12
2	3	Dr. Raj Singh	Orthopedics	8
3	4	Dr. Lisa Park	Dermatology	6
4	5	Dr. Carlos Torres	General Medicine	20

3. Add a column seniority:

years >= 15 → "Principal"

years >= 10 → "Senior"

below 10 → "Associate"

```
# 3. Add a column seniority:
#   years >= 15 → "Principal"
#   years >= 10 → "Senior"
#   below 10   → "Associate"
import numpy as np

if "experience_years" in df.columns:
    conditions = [
        df["experience_years"] >= 15,
        df["experience_years"] >= 10,
        df["experience_years"] < 10
    ]

    choices = ["Principal", "Senior", "Associate"]

    df["Seniority"] = np.select(conditions, choices, default="Unknown")
    print(df)
```

✓ 0.0s

	doctor_id	name	specialty	experience_years	Seniority
0	1	Dr. Anil Mehta	Cardiology	15	Principal
1	2	Dr. Sarah Johnson	Neurology	12	Senior
2	3	Dr. Raj Singh	Orthopedics	8	Associate
3	4	Dr. Lisa Park	Dermatology	6	Associate
4	5	Dr. Carlos Torres	General Medicine	20	Principal

4. Save to processed/doctors_clean.csv

without index.

```
# 4. Save to processed/doctors_clean.csv
# without index.
df.to_csv("processed/doctors_clean.csv", index=False)

df.head()
✓ 0.0s
```

	doctor_id	name	specialty	experience_years	Seniority
0	1	Dr. Anil Mehta	Cardiology	15	Principal
1	2	Dr. Sarah Johnson	Neurology	12	Senior
2	3	Dr. Raj Singh	Orthopedics	8	Associate
3	4	Dr. Lisa Park	Dermatology	6	Associate
4	5	Dr. Carlos Torres	General Medicine	20	Principal

Using function:

```
import pandas as pd
import numpy as np

def clean_doctors():
    df = pd.read_csv("doctors.csv")
    print(df.head())

    df = df.apply(lambda col: col.str.strip() if col.dtype == object else col)
    print("-----")
    if "experience_years" in df.columns:
        conditions = [
            df["experience_years"] >= 15,
            df["experience_years"] >= 10,
            df["experience_years"] < 10
        ]

        choices = ["Principal", "Senior", "Associate"]

        df["Seniority"] = np.select(conditions, choices, default="Unknown")
        print(df.head())

        df.to_csv("processed/doctors_clean.csv", index=False)

        print("\nCleaned file saved to processed/doctors_clean.csv")

if __name__ == "__main__":
    clean_doctors()
```

Task C - Clean appointments.csv:

1. Load raw/appointments.csv.

```
# Task C - Clean appointments.csv:
# 1. Load raw/appointments.csv.
import pandas as pd
import numpy as np

df = pd.read_csv("appointments.csv")
df.head(11)
```

✓ 0.0s

	appt_id	patient_id	doctor_id	appt_date	diagnosis	fee
0	1	1	1	2024-01-10	Hypertension checkup	500
1	2	2	5	2024-01-11	Fever and cold	200
2	3	3	3	2024-01-12	Knee pain evaluation	750
3	4	4	2	2024-01-13	Migraine treatment	600
4	5	5	4	2024-01-14	Acne treatment	300
5	6	1	5	2024-01-15	Follow-up general checkup	200
6	7	6	1	2024-01-16	Cardiac stress test	900
7	8	7	3	2024-01-17	Back pain physiotherapy	400
8	9	8	5	2024-01-18	Routine vaccination	150
9	10	2	2	2024-01-19	Epilepsy management	850

2. Strip whitespace from all string columns.

```
# 2. Strip whitespace from all string columns.
df = df.apply(lambda col: col.str.strip() if col.dtype == object else col)
print(df)
```

✓ 0.0s

	appt_id	patient_id	doctor_id	appt_date	diagnosis	fee
0	1	1	1	2024-01-10	Hypertension checkup	500
1	2	2	5	2024-01-11	Fever and cold	200
2	3	3	3	2024-01-12	Knee pain evaluation	750
3	4	4	2	2024-01-13	Migraine treatment	600
4	5	5	4	2024-01-14	Acne treatment	300
5	6	1	5	2024-01-15	Follow-up general checkup	200
6	7	6	1	2024-01-16	Cardiac stress test	900
7	8	7	3	2024-01-17	Back pain physiotherapy	400
8	9	8	5	2024-01-18	Routine vaccination	150
9	10	2	2	2024-01-19	Epilepsy management	850

3. Convert appt_date to datetime using

pd.to_datetime().

```
# 3. Convert appt_date to datetime using
# pd.to_datetime().
if "appt_date" in df.columns:
    df["appt_date"] = pd.to_datetime(df["appt_date"], errors="coerce")
    print(df)
```

✓ 0.0s

	appt_id	patient_id	doctor_id	appt_date	diagnosis	fee
0	1	1	1	2024-01-10	Hypertension checkup	500
1	2	2	5	2024-01-11	Fever and cold	200
2	3	3	3	2024-01-12	Knee pain evaluation	750
3	4	4	2	2024-01-13	Migraine treatment	600
4	5	5	4	2024-01-14	Acne treatment	300
5	6	1	5	2024-01-15	Follow-up general checkup	200
6	7	6	1	2024-01-16	Cardiac stress test	900
7	8	7	3	2024-01-17	Back pain physiotherapy	400
8	9	8	5	2024-01-18	Routine vaccination	150
9	10	2	2	2024-01-19	Epilepsy management	850

4. Add a column fee_category:

fee >= 700 → "High"

fee >= 400 → "Medium"

fee < 400 → "Low"

```
# 4. Add a column fee_category:
# fee >= 700 → "High"
# fee >= 400 → "Medium"
# fee < 400 → "Low"
if "fee" in df.columns:
    conditions = [
        df["fee"] >= 700,
        df["fee"] >= 400,
        df["fee"] < 400
    ]

    choices = ["High", "Medium", "Low"]
    df["fee_category"] = np.select(conditions, choices, default="Unknown")
    print(df)
```



✓ 0.0s

5. Add a column `day_of_week` extracted from `appt_date`. Hint: `df['appt_date'].dt.day_name()`

```
# 5. Add a column day_of_week extracted from
# appt_date.
if "appt_date" in df.columns:
    df["day_of_week"] = df["appt_date"].dt.day_name()
    print(df)
# Hint: df['appt_date'].dt.day_name()
```

✓ 0.0s

6. Save to `processed/appointments_clean.csv` without index.

```
# 6. Save to processed/appointments_clean.csv
# without index
df.to_csv("processed/appointments_clean.csv")

df.head(11)
```

✓ 0.0s

	appt_id	patient_id	doctor_id	appt_date	diagnosis	fee	fee_category	day_of_week
0	1	1	1	2024-01-10	Hypertension checkup	500	Medium	Wednesday
1	2	2	5	2024-01-11	Fever and cold	200	Low	Thursday
2	3	3	3	2024-01-12	Knee pain evaluation	750	High	Friday
3	4	4	2	2024-01-13	Migraine treatment	600	Medium	Saturday
4	5	5	4	2024-01-14	Acne treatment	300	Low	Sunday
5	6	1	5	2024-01-15	Follow-up general checkup	200	Low	Monday
6	7	6	1	2024-01-16	Cardiac stress test	900	High	Tuesday
7	8	7	3	2024-01-17	Back pain physiotherapy	400	Medium	Wednesday
8	9	8	5	2024-01-18	Routine vaccination	150	Low	Thursday
9	10	2	2	2024-01-19	Epilepsy management	850	High	Friday

STEP 3 - Python: Enrich by merging

Create: processed/enricher.py

1. Load all three processed clean CSV files.

```
#  
# 1. Load all three processed clean CSV files.  
import pandas as pd  
  
patients = pd.read_csv("processed/patients_clean.csv")  
doctors = pd.read_csv("processed/doctors_clean.csv")  
appointments = pd.read_csv("processed/appointments_clean.csv")
```

2. Merge appointments with patients
on patient_id (left join).

```
# 2. Merge appointments with patients  
# on patient_id (left join).  
df = pd.merge(  
    appointments,  
    patients,  
    on="patient_id",  
    how="left"  
)  
print(df.columns)
```

✓ 0.0s

```
Index(['Unnamed: 0', 'appt_id', 'patient_id', 'doctor_id', 'appt_date',  
      'diagnosis', 'fee', 'fee_category', 'day_of_week', 'name', 'age',  
      'blood_type', 'city', 'age_group'],  
      dtype='object')
```

3. Merge the result with doctors
on doctor_id (left join).

```
# 3. Merge the result with doctors  
# on doctor_id (left join).  
df= pd.merge(  
    df,  
    doctors,  
    on="doctor_id",  
    how="left"  
)  
print(df.columns)
```

✓ 0.0s

```
Index(['Unnamed: 0', 'appt_id', 'patient_id', 'doctor_id', 'appt_date',  
      'diagnosis', 'fee', 'fee_category', 'day_of_week', 'name_x', 'age',  
      'blood_type', 'city', 'age_group', 'name_y', 'specialty',  
      'experience_years', 'Seniority'],  
      dtype='object')
```

4. The final merged DataFrame should have:

appt_id, patient name, patient age_group,
patient city, doctor name, specialty,
seniority, appt_date, day_of_week,
diagnosis, fee, fee_category.

5. Select only these meaningful columns

and rename them cleanly if needed.

```
# 5. Select only these meaningful columns
# and rename them cleanly if needed.
df_final = df[[
    'appt_id',
    'appt_date',
    'patient_id',
    'name_x',
    'age',
    'city',
    'doctor_id',
    'name_y',
    'specialty',
    'experience_years',
    'diagnosis',
    'fee',
    'fee_category',
    'day_of_week'
]].rename(columns={
    'name_x': 'patient_name',
    'name_y': 'doctor_name'
})
print(df_final.columns)
```

✓ 0.0s

```
Index(['appt_id', 'appt_date', 'patient_id', 'patient_name', 'age', 'city',
      'doctor_id', 'doctor_name', 'specialty', 'experience_years',
      'diagnosis', 'fee', 'fee_category', 'day_of_week'],
      dtype='object')
```

6. Save to processed/enriched_data.csv

without index.

```
# 6. Save to processed/enriched_data.csv
# without index.
df_final.to_csv("processed/enriched_data.csv")

# 7. Print shape and first 5 rows to verify.
print("Shape:", df_final.shape)
```

✓ 0.0s

Shape: (10, 14)

7. Print shape and first 5 rows to verify.

df.head(10)

0.0s

Python

Unnamed: 0	appt_id	patient_id	doctor_id	appt_date	diagnosis	fee	fee_category	day_of_week	name_x	age
0	0	1	1	2024-01-10	Hypertension checkup	500	Medium	Wednesday	Raj Kumar	34
1	1	2	2	2024-01-11	Fever and cold	200	Low	Thursday	Sneha Patel	39
2	2	3	3	2024-01-12	Knee pain evaluation	750	High	Friday	Tom Williams	46
3	3	4	4	2024-01-13	Migraine treatment	600	Medium	Saturday	Fatima Al-Sayed	29
4	4	5	5	2024-01-14	Acne treatment	300	Low	Sunday	Liu Yang	24
5	5	6	1	2024-01-15	Follow-up general checkup	200	Low	Monday	Raj Kumar	34
6	6	7	6	2024-01-16	Cardiac stress test	900	High	Tuesday	Ana Gonzalez	56
7	7	8	7	2024-01-17	Back pain physiotherapy	400	Medium	Wednesday	James Okafor	32

STEP 4 - Linux verify

1. Check row count of enriched_data.csv.

Should match appointments (10 rows + header).

2. Display the header of enriched_data.csv.

3. Search for all appointments handled

by General Medicine specialty.

grep "General Medicine" processed/enriched_data.csv

```
prashant@DESKTOP-B9KE2N7:/mnt/d/kode-x notes/python/python_exercise's/day6_project/hospital_pipeline/raw$ cd processed/
prashant@DESKTOP-B9KE2N7:/mnt/d/kode-x notes/python/python_exercise's/day6_project/hospital_pipeline/raw/processed$ cat enriched_data
.csv | wc -l
11
prashant@DESKTOP-B9KE2N7:/mnt/d/kode-x notes/python/python_exercise's/day6_project/hospital_pipeline/raw/processed$ head -n 1 enriched_data.csv
,appt_id,appt_date,patient_id,patient_name,age,city,doctor_id,doctor_name,specialty,experience_years,diagnosis,fee,fee_category,day_of_week
prashant@DESKTOP-B9KE2N7:/mnt/d/kode-x notes/python/python_exercise's/day6_project/hospital_pipeline/raw/processed$ grep -i "General Medicine" enriched_data.csv
1,2,2024-01-11,2,Sneha Patel,39,Mumbai,5,Dr. Carlos Torres,General Medicine,20,Fever and cold,200,Low,Thursday
5,6,2024-01-15,1,Raj Kumar,34,Delhi,5,Dr. Carlos Torres,General Medicine,20,Follow-up general checkup,200,Low,Monday
8,9,2024-01-18,8,Emily Carter,19,Toronto,5,Dr. Carlos Torres,General Medicine,20,Routine vaccination,150,Low,Thursday
prashant@DESKTOP-B9KE2N7:/mnt/d/kode-x notes/python/python_exercise's/day6_project/hospital_pipeline/raw/processed$ cat enriched_data
```